Serverless Design

in Cloud

Yegor Bugayenko

Lecture #12 out of 16
80 minutes

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Cloud
On-premise

- Amazon Web Services
- Docker Containers
- PaaS, IaaS, SaaS, EaaS, etc.
- P2P: BitTorrent, Blockchain, and Beyond
- Books, Venues, Call-to-Action

Chapter #1:

Amazon Web Services



S3]Simple Storage Service (S3)

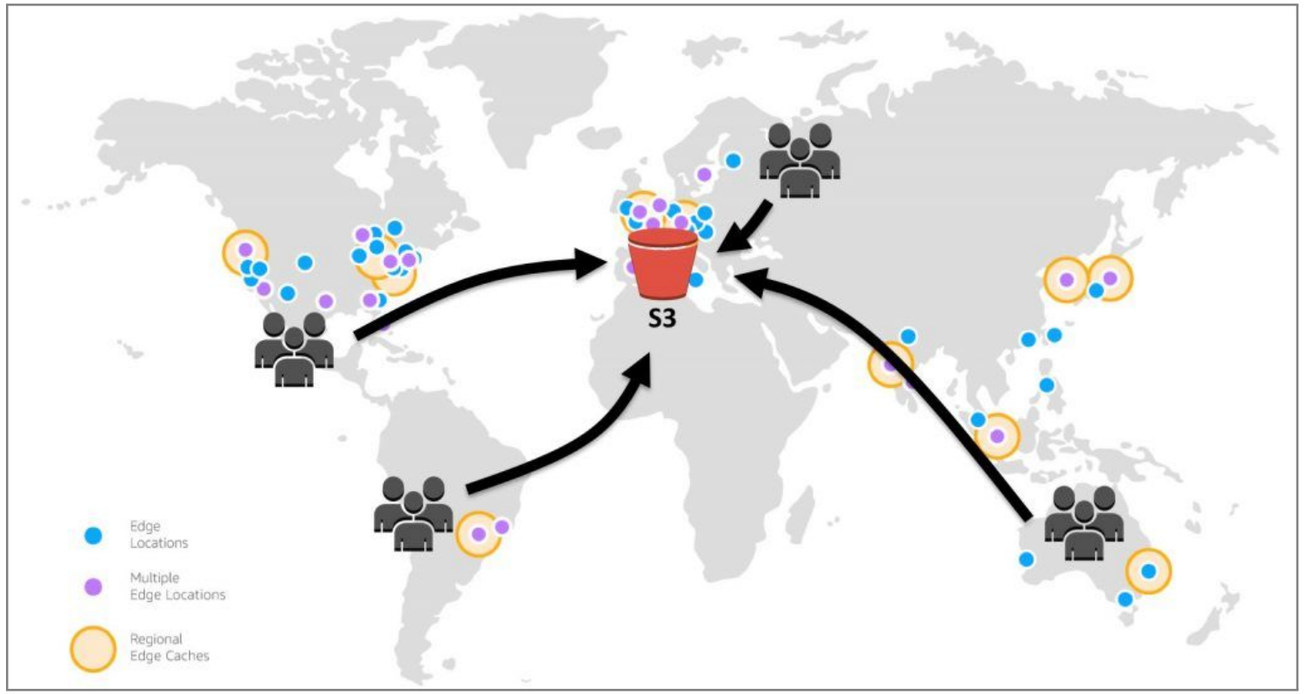
```ruby
def object_uploaded?(s3_client, bucket_name, object_key)
  response = s3_client.put_object(
    bucket: bucket_name,
    key: object_key
  )
  if response.etag
    return true
  else
    return false
  end
rescue StandardError => e
  puts "Error uploading object: #{e.message}"
  return false
end
```



EC2]Elastic Compute Cloud (EC2)

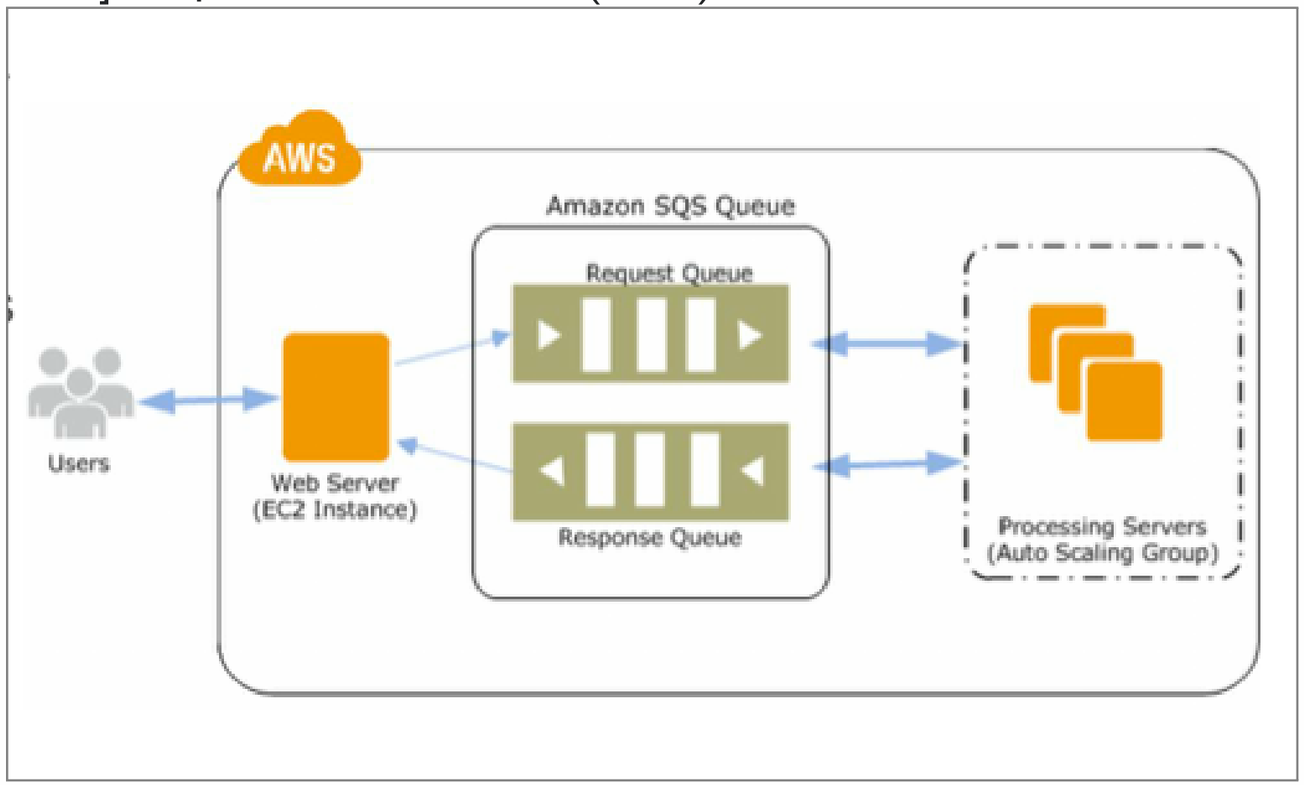
```
instance = ec2_resource.create_instances(
  image_id: image_id,
  min_count: 1,
  max_count: 1,
  key_name: key_pair_name,
  instance_type: instance_type,
  user_data: encoded_script
)
```

CloudFront

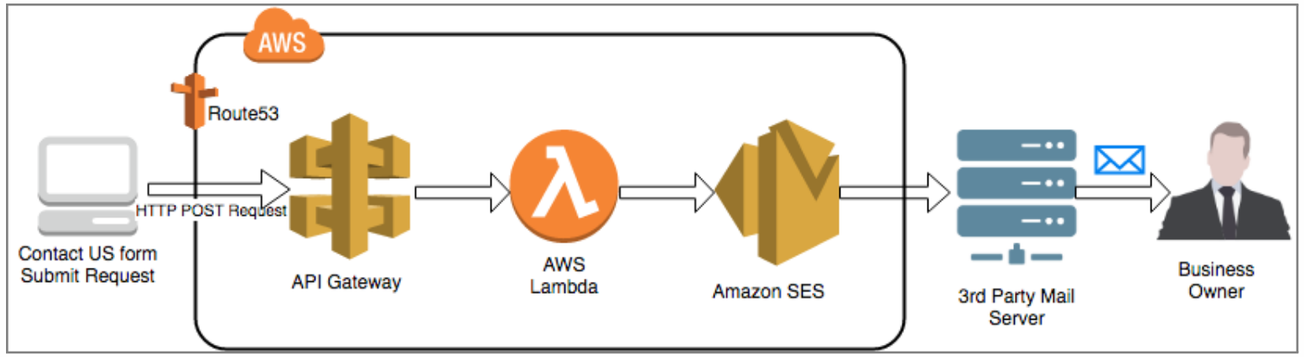

S3
Edge Locations
Multiple Edge Locations
Regional Edge Caches

[

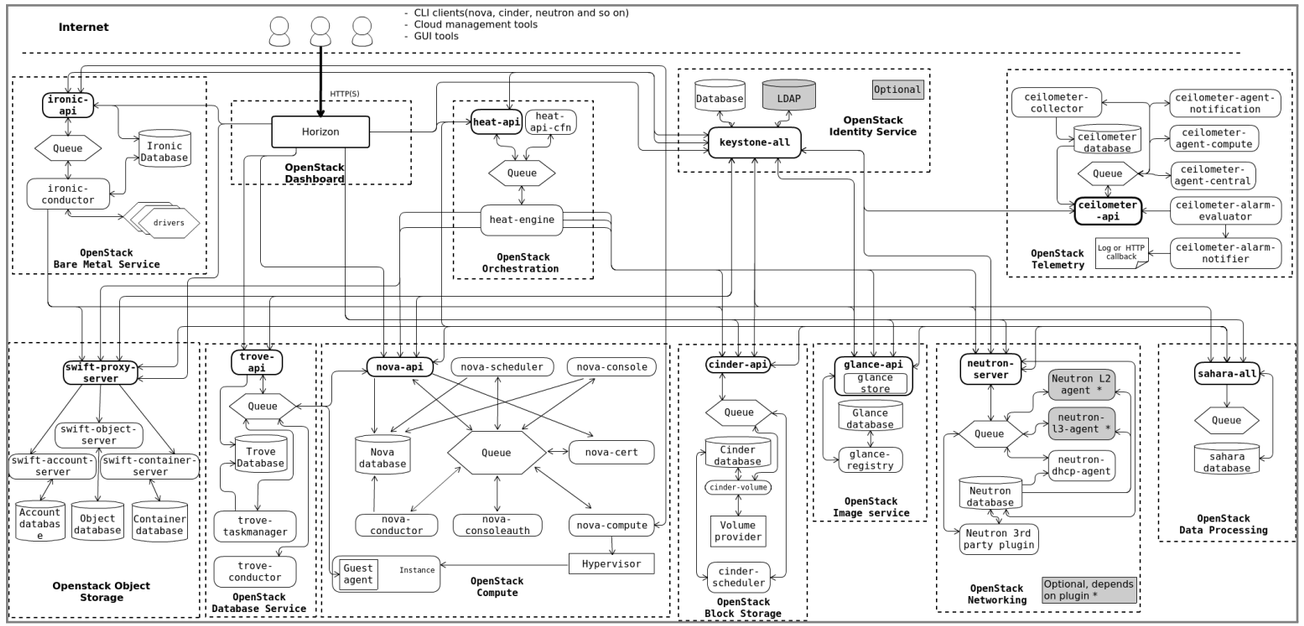
SQS]Simple Queue Service (SQS)



Lambda


AWS
Route53
HTTP POST Request
Contact US form
Submit Request
API Gateway
AWS
Lambda
Amazon SES
3rd Party Mail
Server
Business
Owner

OpenStack



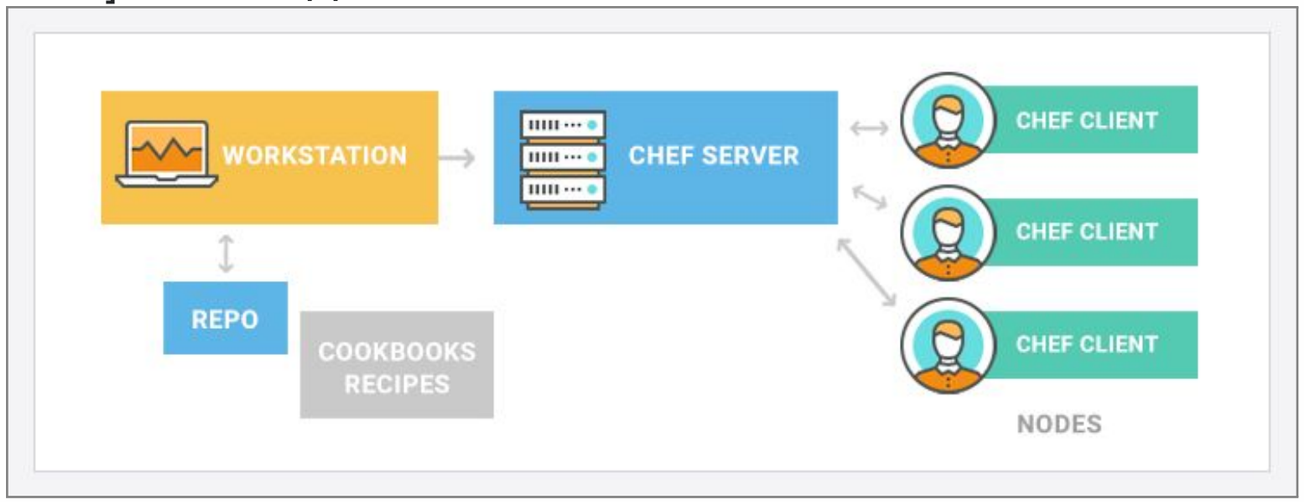
https://docs.openstack.org/install-guide/get-started-logical-architecture.html

Chapter #2:

Docker Containers

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Chef]Chef, Puppet, etc.



Dockerfile

```
FROM yegor256/rultor-image
MAINTAINER Yegor Bugayenko <yegor256@gmail.com>
LABEL Description="yegor256.com" Vendor="Yegor Bugayenko" Version="1.0"

ENV DEBIAN_FRONTEND=noninteractive

RUN sudo apt-get -y update --fix-missing
RUN sudo apt-get -y install aspell aspell-en
RUN sudo apt-get -y install graphviz
RUN sudo apt-get -y update --fix-missing
RUN sudo apt-get -y install gnuplot
RUN sudo apt-get -y install s3cmd
RUN sudo apt-get -y install fontforge liblapack-dev

RUN sudo apt-get -y install plantuml
COPY plantuml.jar /usr/share/plantuml/plantuml.jar

RUN wget --no-check-certificate --quiet https://cmake.org/files/v3.5/cmake-3.5.1.tar.gz && \
  tar xf cmake-3.5.1.tar.gz && \
  rm -rf _cmake-3.5.1 && \
  mv cmake-3.5.1 _cmake-3.5.1 && \
  cd _cmake-3.5.1 && \
  ./configure && \
  make && \
  make install

RUN git clone https://github.com/htacg/tidy-html5.git _tidy-html5 && \
  cd _tidy-html5/build/cmake && \
  git checkout 5.1.25 && \
  cmake ../.. && \
  make && \
  make install

RUN sudo apt-get install -y woff2
COPY woff.zip /tmp/woff.zip
RUN unzip /tmp/woff.zip -d _sfnt2woff && \
  cd _sfnt2woff && \
  make && \
  cp sfnt2woff /usr/local/bin/

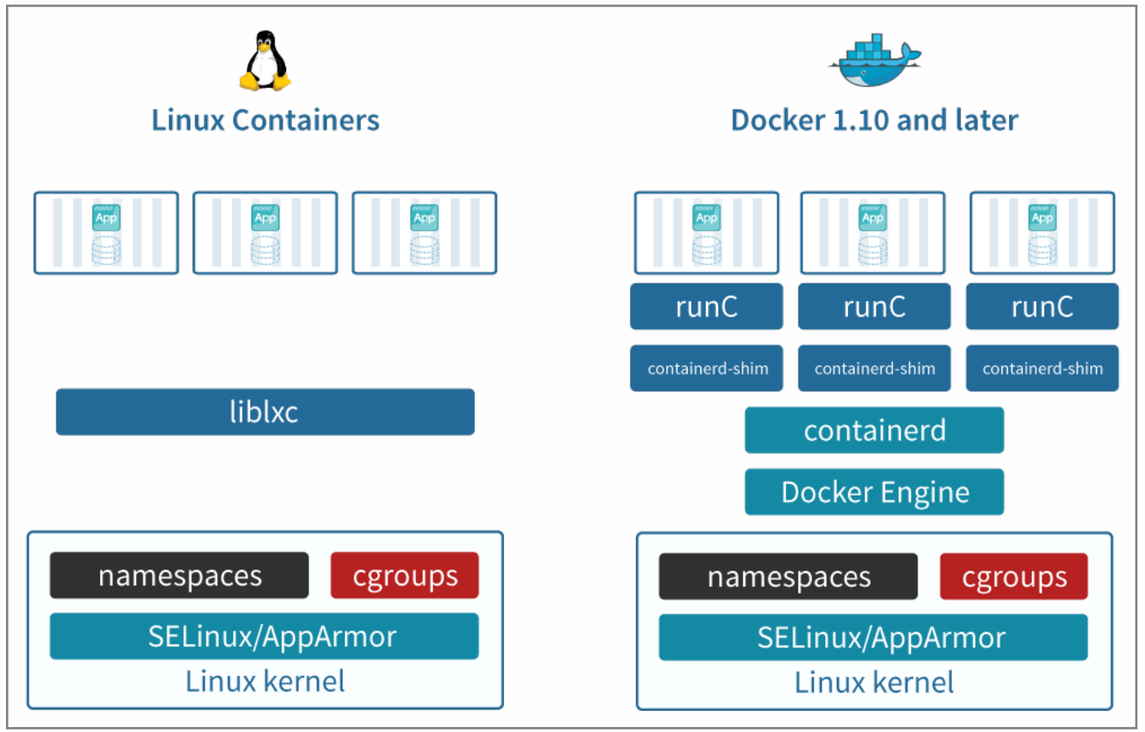
RUN sudo apt-get -y update --fix-missing
RUN sudo apt-get install -y libxml2-utils

RUN npm install -g cssshrink@0.0.5

RUN /bin/bash -l -c "gem install jgd -v 1.6.5"
```

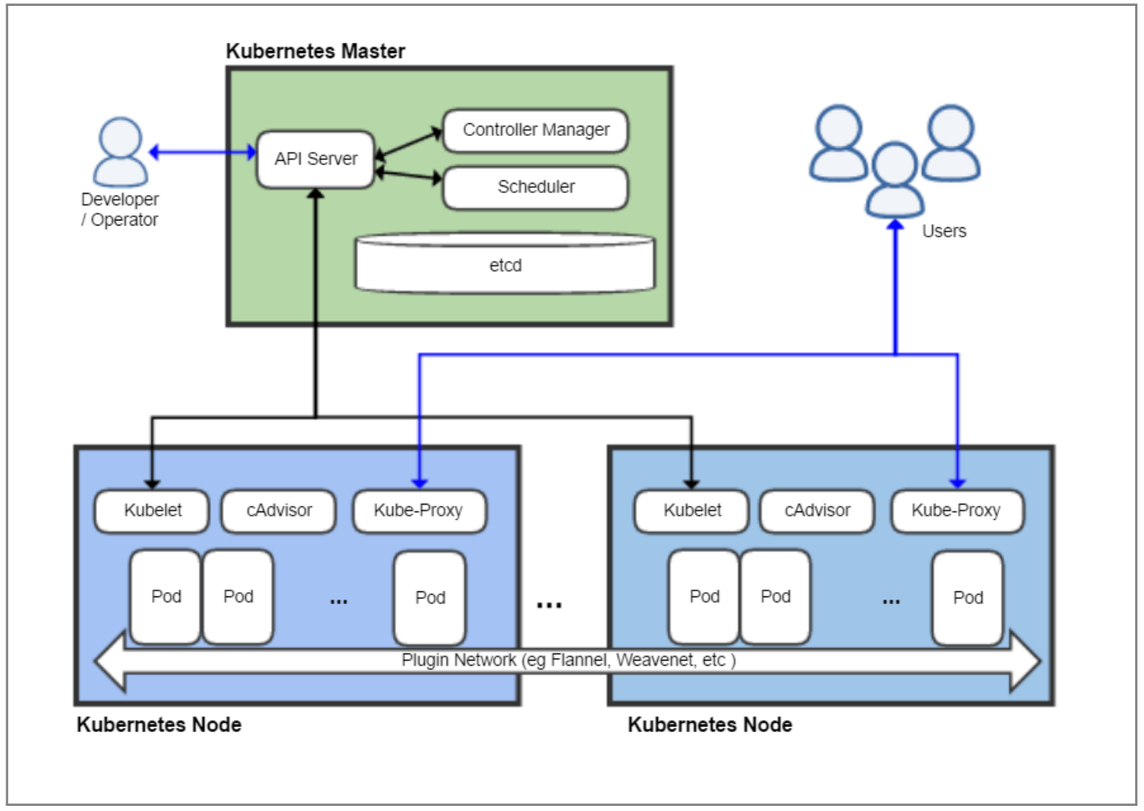
```
RUN git clone https://github.com/htacg/tidy-html5.git _tidy-html5 && \
  cd _tidy-html5/build/cmake && \
  git checkout 5.1.25 && \
  cmake ../.. && \
  make && \
  make install
```

LXC


Linux Containers
App
App
App
liblxc
namespaces
cgroups
SELinux/AppArmor
Linux kernel
Docker 1.10 and later
App
App
App
runC
runC
runC
containerd-shim
containerd-shim
containerd-shim
containerd
Docker Engine
namespaces
cgroups
SELinux/AppArmor
Linux kernel

LXC is an operating-system-level virtualization method for running multiple isolated Linux systems on a control host using a single Linux kernel.





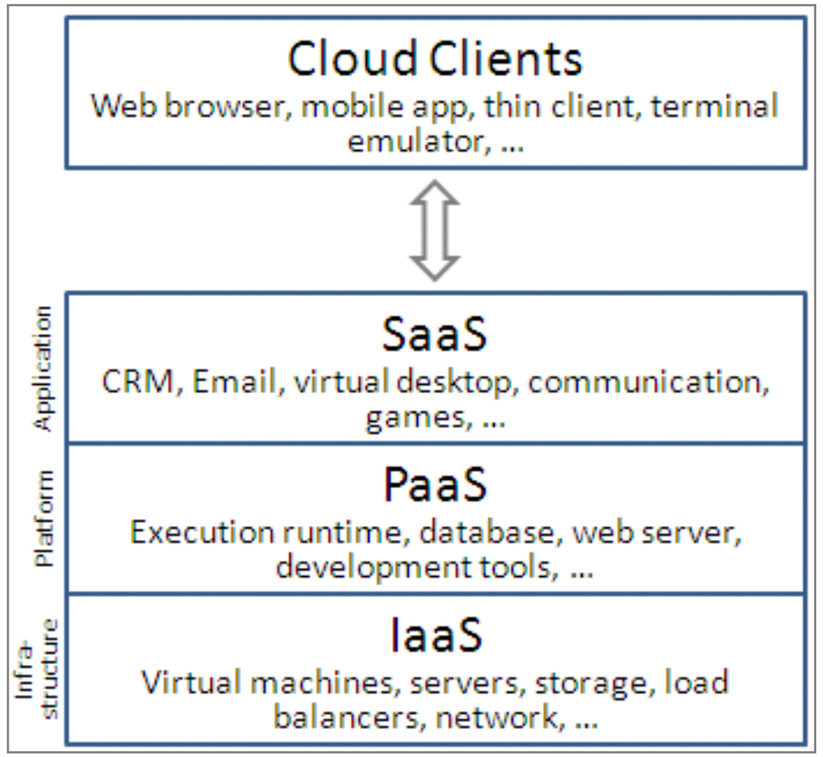
K8s]Kubernetes

Chapter #3:

PaaS, IaaS, SaaS, EaaS, etc.

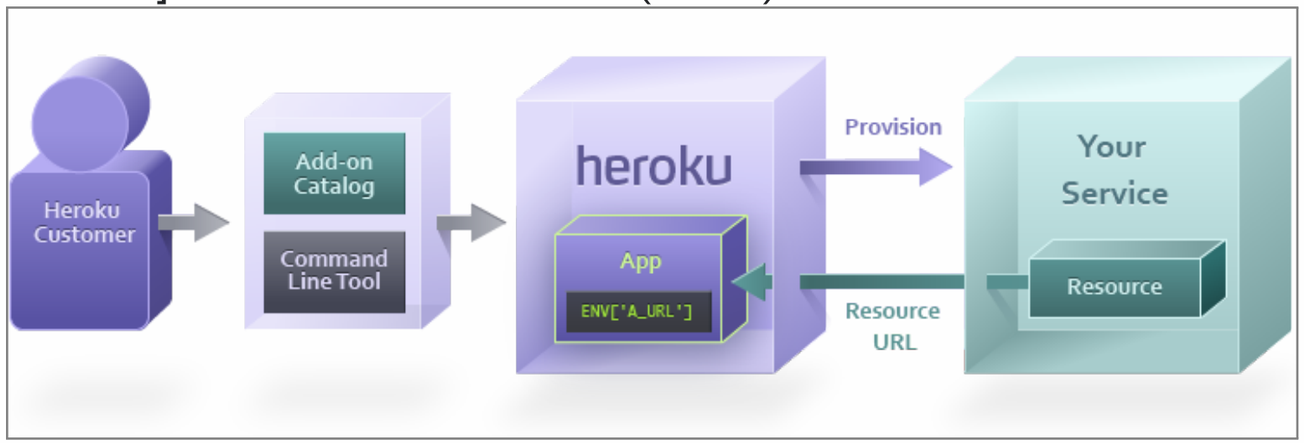
[

EaaS]Everything as a Service (EaaS)



[

Heroku]Platform as a Service (PaaS)





https://martinfowler.com/articles/serverless.html

Serverless]Serverless Architecture Serverless architectures are application designs that incorporate third-party “Backend as a Service” (BaaS) services, and/or that include custom code run in managed, ephemeral containers on a “Functions as a Service” (FaaS) platform.

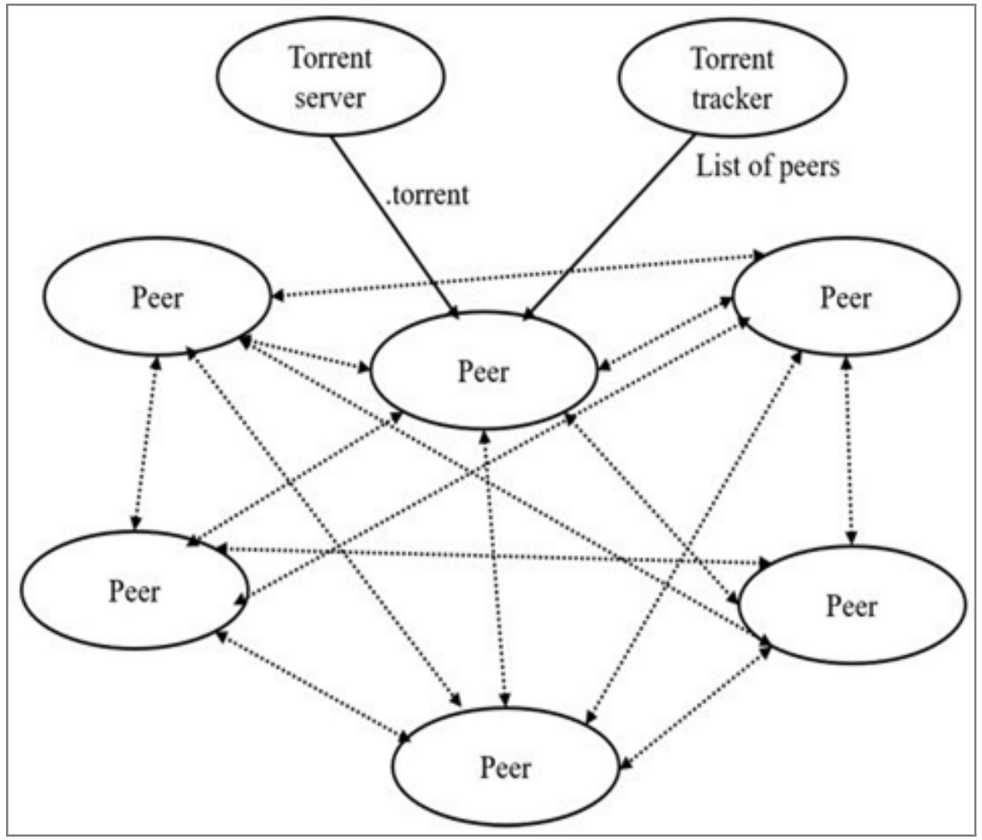
Fundamentally, FaaS is about running backend code without managing your own server systems or your own long-lived server applications.

AWS RDS vs. AWS DynamoDB

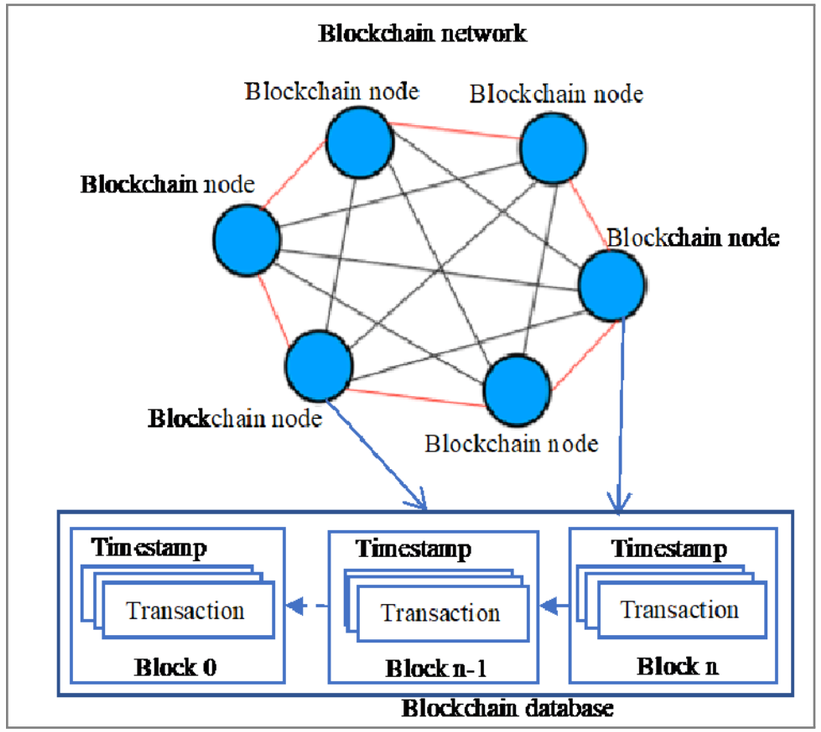
Chapter #4:

P2P: BitTorrent, Blockchain, and Beyond

BitTorrent

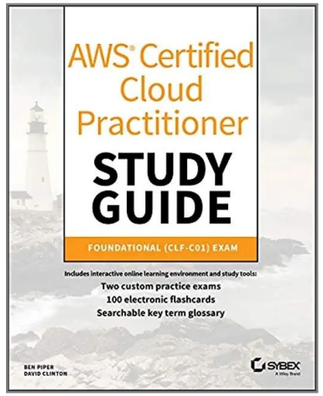

Torrent server
Torrent tracker
.torrent
List of peers
Peer
Peer
Peer
Peer
Peer
Peer

Blockchain

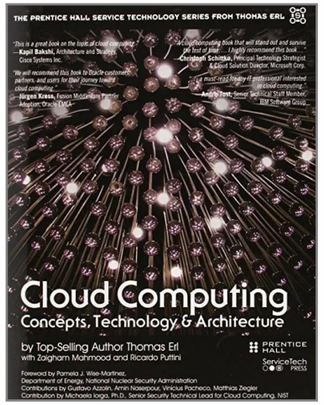

Blockchain network
Blockchain node
Blockchain node
Blockchain node
Blockchain node
Blockchain node
Blockchain node
Timestamp
Transaction
Block 0
Timestamp
Transaction
Block n-1
Timestamp
Transaction
Block n
Blockchain database

Chapter #5:

Books, Venues, Call-to-Action



Ben Piper and David Clinton. *AWS Certified Cloud Practitioner Study Guide: CLF-C01 Exam*. Wiley, 2019



Thomas Erl, Ricardo Puttini, and Zaigham Mahmood. *Cloud Computing: Concepts, Technology & Architecture*. Prentice Hall, 2013

Where to go:

AWS Certified Solution Architect

Call to Action:

Use two AWS services in your app.

Still unresolved issues:

- How to simplify serverless design?
- How to manage data serverless?
- How to decentralize better than Blockchain?
- How to enable decentralized computing on IoT devices?

References

[1] Thomas Erl, Ricardo Puttini, and Zaigham Mahmood. *Cloud Computing: Concepts, Technology & Architecture*. Prentice Hall, 2013.

[2] Ben Piper and David Clinton. *AWS Certified Cloud Practitioner Study Guide: CLF-C01 Exam*. Wiley, 2019.